

ALEXIS MARCHAND

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EMPLOYMENT

<i>from</i> 9/2025 <i>to</i> ...	IM PAN, Warszawa (Poland) Adiunkt (Assistant Professor) Research project: <i>Analysis on groups</i> (NCN Maestro 13) Mentor: PIOTR NOWAK
<i>from</i> 5/2026 <i>to</i> 7/2026	Universität Regensburg / Humboldt-Stiftung (Germany) Humboldt Research Fellow Research project: <i>Invariants of groups and non-positive curvature</i> Mentor: CLARA LÖH
<i>from</i> 9/2024 <i>to</i> 9/2025	Kyōto University / JSPS (Japan) JSPS Postdoctoral Fellow Research project: <i>Stable commutator length in finitely presented groups</i> Mentor: KŌJI FUJIWARA

EDUCATION

<i>from</i> 10/2021 <i>to</i> 10/2024	University of Cambridge (U.K.) • <i>Ph.D. in Pure Mathematics</i> Thesis: <i>Computations and lower bounds for scl and the relative Gromov seminorm</i> Defended on: 15/7/2024 Funding: EPSRC PhD studentship Supervisor: HENRY WILTON
<i>from</i> 9/2017 <i>to</i> 8/2021	École Normale Supérieure de Lyon (France) • <i>Dipl. ENS</i> Funding: ENS studentship awarded by national competitive examination (rank: 16)
<i>from</i> 9/2020 <i>to</i> 7/2021	Université Grenoble Alpes (France) • <i>M.Sc. in Mathematics</i> Mention Très Bien Thesis: <i>Automorphismes extérieurs de produits libres : revêtements abéliens caractéristiques et représentations libres</i> Supervisor: FRANÇOIS DAHMANI
<i>from</i> 10/2019 <i>to</i> 6/2020	University of Cambridge (U.K.) • <i>M.A.St. in Mathematics</i> Pass with Merit Essay: <i>Rigidity theorems for hyperbolic groups</i> Supervisor: HENRY WILTON
<i>from</i> 9/2017 <i>to</i> 7/2019	ÉNS de Lyon / Université Lyon 1 (France) • <i>B.Sc. in Mathematics</i> First two years of ENS curriculum in mathematics (L3+M1)

from 9/2015 | **Lycée Louis le Grand, Paris** (France) • *Preparatory class*
to 7/2017 | Mathematics, physics, computer science

from 9/2012 | **Lycée Gustave Flaubert, Rouen** (France) • *Baccalauréat*
to 7/2015 | Scientific pathway, British international option

PUBLICATIONS (PEER-REVIEWED)

- [1] Alexis Marchand, From letter-quasimorphisms to angle structures and spectral gaps for scl. **Israel J. Math.** (online first). DOI:[10.1007/s11856-026-2898-3](https://doi.org/10.1007/s11856-026-2898-3) • arXiv:[2402.13856](https://arxiv.org/abs/2402.13856)
- [2] Alexis Marchand, Bavard duality for the relative Gromov seminorm, **Enseign. Math.** **72** (2026), no. 1-2, 1-46. DOI:[10.4171/lem/1087](https://doi.org/10.4171/lem/1087) • arXiv:[2305.02149](https://arxiv.org/abs/2305.02149)
- [3] Alexis Marchand, Isometric embeddings of surfaces for scl, **Groups Geom. Dyn.** **19** (2025), no. 4, 1445-1478. DOI:[10.4171/ggd/845](https://doi.org/10.4171/ggd/845) • arXiv:[2302.04133](https://arxiv.org/abs/2302.04133)
- [4] Rémi Coulon and Alexis Marchand, Growth and co-growth in negatively curved groups, in Dahmani, Erschler, Horbez, Wise (eds.), Groups acting on fractals, hyperbolicity and auto-similarity, **Panoramas et Synthèses** **63** (2025).
- [5] Alexis Marchand, Free representations of outer automorphism groups of free products via characteristic abelian coverings, **J. Group Theory** **26** (2023), no. 2, 399-420. DOI:[10.1515/jgth-2021-0154](https://doi.org/10.1515/jgth-2021-0154) • arXiv:[2107.11230](https://arxiv.org/abs/2107.11230)

EXPOSITORY ARTICLES

- [6] Alexis Marchand, Makers of patterns: from Escher to Coxeter, **Eureka** (2020).
- [7] Alexis Marchand, Géométrie des groupes et courbure négative, **Journal de Mathématiques des Élèves de l'ENS de Lyon** (2020).

GRANTS & FUNDING

from 9/2024 | **JSPS** • Fellow's Research Support Allowance
to 9/2025 | Allocated amount: JPY 840,000
Topic: *Stable commutator length in finitely presented groups*

6/2025 | **RIMS, Kyōto University** • RIMS Workshop (type A)
Allocated amount: JPY 350,000
Topic: *Spectrum problems for scl via S-machines*

AWARDS

3/2023 | **Smith-Knight & Rayleigh-Knight prize** • University of Cambridge
Grade 3 (GBP 300), for the essay *Isometric embeddings of surfaces for scl*

RESEARCH VISITS

11/2025 | **INI, Cambridge** (U.K.) • Three-week visit
Programme: *Operators, graphs, groups* (participant)

- 6/2025 | **RIMS, Kyōto University** (Japan) • Two-week small-group workshop
Programme: *Spectrum problems for scl via S-machines* (co-organiser)
- 6/2025 | **KIAS, Seoul** (South Korea) • One-week visit • invited by SANG-HYUN KIM
- 4/2025 | **University of Hawai'i at Mānoa** (U.S.) • Ten-day visit • invited by YASH LODHA
- 4/2022 | **IHP, Paris** (France) • One two-week visit and one three-week visit
and 6/2022 | Programme: *Groups acting on fractals* (participant)
- from 6/2019 | **IMPA, Rio de Janeiro** (Brazil) • Two-month M.Sc. internship
to 8/2019 | Topic: *Topology of complex affine varieties* • Supervisor: HOSSEIN MOVASATI
- from 5/2018 | **KIT, Karlsruhe** (Germany) • Six-week B.Sc. internship
to 6/2018 | Topic: *Geometry of Coxeter groups* • Supervisor: CATERINA CAMPAGNOLO

HIGHER EDUCATION TEACHING EXPERIENCE

- from 1/2023 | **Geometry** • Trinity College & Peterhouse College, Cambridge
to 5/2024 | Part IB supervisions • 34 contact hours • in English
- from 1/2024 | **Geometric Group Theory** • University of Cambridge
to 3/2024 | Part III examples classes • 8 contact hours • in English
- from 10/2022 | **Analysis & Topology** • Trinity College & Peterhouse College, Cambridge
to 1/2024 | Part IB supervisions • 46 contact hours • in English
- from 12/2022 | **Admissions interviews** • Christ's College, Cambridge
to 12/2023 | Co-interviewer for undergraduate admissions in mathematics • in English
- from 1/2023 | **Group Cohomology** • University of Cambridge
to 5/2023 | Part III drop-in sessions • 4 contact hours • in English
- from 1/2022 | **Riemann Surfaces** • Trinity College & Peterhouse College, Cambridge
to 5/2022 | Part II supervisions • 31 contact hours • in English
- from 10/2021 | **Linear Algebra** • Trinity College, Cambridge
to 1/2022 | Part IB supervisions • 16 contact hours • in English
- from 10/2021 | **Mapping Class Groups** • University of Cambridge
to 1/2022 | Part III drop-in sessions • 4 contact hours • in English
- from 9/2018 | **Colles de Mathématiques** • Lycée Aux Lazaristes, Lyon
to 5/2019 | Oral examinations in mathematics for first-year students in a preparatory class • 44 contact hours • in French

VOLUNTEER TEACHING

- 3/2025 | **JSPS science dialogue** • Koshi High School, Fukui
Lecture in English and Japanese for a high school science club
- from 10/2023 | **High school project supervision** • Rusanivskyi Lyceum, Kyiv (online)
to 4/2024 | Student: OLHA MYKHAILYK • Topic: *Inversions and Möbius transformations*
Second prize at the city stage in the Junior Academy of Sciences of Ukraine research competition

from 10/2019 to 3/2020	Tutoring • Coleridge Community College & St Faith's School, Cambridge Volunteer tutoring in mathematics in secondary schools, under the <i>Stimulus</i> programme
from 9/2017 to 6/2020	French as a foreign language • ItinérENS, Lyon French classes for refugees • more than 100 contact hours

SERVICE

Refereeing for *Trans. Amer. Math. Soc.*, *Algebr. Geom. Topol.*, *KIAS Springer Ser. Math.*

from 9/2025 to ...	Analytic & geometric group theory seminar • IM PAN, Warszawa Co-organiser
9/2025	World of GroupCraft V • Online Co-organiser
6/2025	RIMS Workshop • RIMS, Kyōto University Topic: <i>Spectrum problems for scl via S-machines</i> • Co-organiser
from 2/2025 to 4/2025	Reading seminar • Kyōto University Topic: L^2 -Betti numbers • Organiser
from 9/2023 to 8/2024	Junior geometry seminar • University of Cambridge Co-organiser
from 8/2023 to 5/2024	MCR committee • Trinity College, Cambridge MCR (Middle Combination Room) committee member in charge of: the MCR's website, mailing list, elections, lunchtime seminars, etc.
from 10/2022 to 12/2022	Reading seminar • University of Cambridge Topic: <i>Generalisations of hyperbolic groups</i> • Organiser

INVITED TALKS

6/2026	Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future • ICMAT, Madrid
6/2026	Séminaire géométrie, algèbre, dynamique et topologie • Université Bourgogne Europe
6/2026	Séminaire de géométrie différentielle • Université de Lorraine
5/2026	LKS-Seminar (Oberseminar für Gruppen und geometrische Topologie) • Universität Regensburg
11/2025	Algebra & geometry seminar • Newcastle University
11/2025	Geometric group theory seminar • University of Cambridge
9/2025	Analytic & geometric group theory seminar • IM PAN, Warszawa
7/2025	Tuesday seminar on topology • University of Tōkyō
7/2025	Geometry seminar • Tōhoku University
6/2025	Topology seminar • KIAS, Seoul
5/2025	Differential topology seminar • Kyōto University

- 4/2025 | Mathematics department colloquium • **University of Hawai'i at Mānoa**
- 4/2025 | Topology seminar • **Kanazawa University**
- 11/2024 | First MMS workshop for young researchers • **Kyōto University**
- 11/2024 | Séminaire virtuel francophone groupes et géométrie • **Online**
- 5/2024 | Seminari dei baby geometri • **Università di Pisa / Scuola Normale Superiore**
- 4/2024 | International young seminar on bounded cohomology & simplicial volume • **Online**
- 4/2024 | Geometry and geometric analysis seminar • **Purdue University (online)**
- 3/2024 | Geometric group theory seminar • **University of Oxford**
- 12/2023 | Bristol junior geometry seminar • **University of Bristol**
- 6/2023 | International young seminar on bounded cohomology & simplicial volume • **Online**
- 4/2023 | Methusalem junior seminar • **KU Leuven**
- 3/2023 | Geometry graduate colloquium • **ETH Zürich**
- 2/2023 | Junior topology & group theory seminar • **University of Oxford**
- 2/2023 | Geometric group theory seminar • **University of Cambridge**
- 6/2022 | Séminaire compréhensible • **Université Grenoble Alpes**
- 4/2022 | Séminaire des doctorants de St-Charles • **Aix-Marseille Université**

CONFERENCES ATTENDED

- 1/2026 | Perspectives in ergodic theory, randomness, and geometry • **Uniwersytet Jagielloński, Kraków**
- 11/2025 | Stability and probabilistic methods • **INI, Cambridge**
- 2/2025 | 20th East Asian conference on geometric topology • **University of Tōkyō**
Contributed talk
- 1/2025 | Geometry and dynamics in low dimensions • **VAST, Hanoi**
Contributed talk
- 4/2024 | Groups, topology, & applications Gran Bilbao • **UPV / EHU, Bilbao**
- 3/2024 | Topological & homological methods in group theory • **Universität Bielefeld**
- 2/2024 | GAGTA • **CIRM, Marseille**
Contributed talk & poster
- 7/2023 | Postgraduate group theory conference • **Heriot-Watt University, Edinburgh**
Contributed talk
- 5/2023 | Geometry of subgroups • **CRM, Montréal**
- 1/2023 | LMS regional meeting • **University of Southampton**
Contributed poster
- 9/2022 | Recent advances in bounded cohomology • **Universität Regensburg**
- 6/2022 | Hyperbolic groups and their generalisations • **IHP, Paris**

4/2022 | Mapping class groups and Out (F_n) • **IHP, Paris**

4/2022 | Groups acting on fractals • **CIRM, Marseille**
 Note-taker for a minicourse [4]

2/2022 | Workshop on finiteness properties • **KIT, Karlsruhe**

LANGUAGES

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| • French • Native language | • German • Advanced (B2) |
| • English • Fluent (C2)
Passed the CAE (Cambridge) | • Portuguese • Advanced (B2) |
| • Japanese • Advanced (B2)
Passed the JLPT N2 | • Russian • Intermediate (A2/B1) |
| | • Polish • Intermediate (A2/B1) |

June 4, 2026